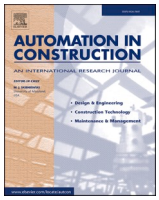
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Real-time bridge disaster management: Enabling technology and application framework

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ABSTRACT

Bridges are susceptible to severe damage from natural disasters, heavy traffic loads, and material degradation, necessitating timely and accurate information for effective emergency response. Current bridge disaster management systems often fail to meet real-time requirements due to interoperability challenges and fragmented functionalities across different phases. This paper systematically reviews 146 research articles on bridge disaster management, summarising the key challenges and potential disasters that bridges may face during their operational lifespan and highlighting the technological applications and decision-making requirements for effective disaster management. Key limitations in existing systems include inadequate technology integration, delayed response times, and insufficient coordination. To address these gaps, this paper proposes a Real-time Bridge Disaster Management (RtBDM) framework, which integrates advanced digital technologies to enable real-time monitoring, data analysis, and adaptive decision-making. The proposed framework offers practical solutions to enhance bridge resilience and safety during disasters and provides valuable insights for future research in this field.

1. Introduction

As critical components of transportation infrastructure, bridges are highly susceptible to damage due to external or internal factors such as natural disasters, heavy traffic loads, or material degradation, which can result in serious casualties and economic losses [1]. Historical disasters, including the 2008 Wenchuan earthquake, the 2023 Pazarcik earthquake, and the 2020 Medicane Ianos, as well as the flood in the Western Black Sea Basin of Turkey in 2021, have severely disrupted regional highway transportation systems, highlighting the vulnerability of bridges [2–5]. From 1996 to 2017, typhoons caused 83 % of Taiwan's bridges to collapse [6]. In India, the major cause of damage to bridges is natural disasters, with 80.30 %, of which hydraulic-induced causes 51.19 %. Another major cause is the deterioration of materials (10.10 %), followed by design and construction (4.13 %), overloading (3.28 %) and human factors at 2.19 % [7]. These incidents underscore the urgent need for comprehensive disaster management strategies for bridges, encompassing pre-disaster warning, emergency response, and post-disaster recovery [8].

Despite significant advancements in disaster prevention and mitigation technologies, such as satellites, drones, and sensor networks, their integration and implementation across different disaster management phases of bridges remains fragmented. This fragmentation limits the acquisition, transmission, and management of real-time data, which is essential for informed decision-making during disasters. Current systems often fail to provide the real-time, dynamic information needed for effective disaster management, resulting in delays in response and recovery. Unlike other infrastructures, such as urban buildings or water networks, bridges face unique challenges in disaster management due to their critical role in maintaining connectivity [9]. For instance, flooding and earthquakes can compromise bridge stability and functionality, leading to severe consequences for rescue operations and regional mobility. These challenges highlight the need for a unified real-time disaster management system tailored specifically to bridge infrastructure.

The digital transformation of infrastructure management and the adoption of emerging technologies have opened new avenues for real-time disaster management [10]. Among these technologies, Building Information Modelling (BIM) and Digital Twin (DT) have demonstrated significant potential. BIM serves as a management platform for various

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0926-5805/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

types of information and knowledge, enabling the creation and management of digital 3D models of structures in a standardised and digitised manner [11]. On the other hand, DT mirrors the real-time state of a structure into a virtual twin model, utilising real-time monitoring data for continuous updating and analysis, which then feeds back into the BIM model for design optimisation [12]. While these technologies have shown utility across diverse infrastructures, bridges require tailored applications or methods to address specific issues, such as real-time structural integrity assessment under dynamic loads or environmental stresses.

This paper aims to address these gaps by pursuing the following objectives:

1) To identify and categorise the types of disasters that bridges may encounter during their operational lifespan.
2) To evaluate the applicability and limitations of current real-time disaster management technologies for bridges.
3) To propose a comprehensive Real-time Bridge Disaster Management (RtBDM) framework that integrates BIM and DT technologies for enhanced timeliness, coordination, and efficiency across all disaster phases.

The rapid development of information and communication technologies (ICT), such as the Internet of Things (IoT), artificial intelligence (AI), data analytics (DA), laser scanning (LS), machine learning (ML), and virtual and augmented reality (VR and AR), has made it possible to gather disaster-related data for real-time disaster assessment, emergency response, and post-disaster recovery [13–16]. For instance, Ford and Wolf [15] developed a smart city model based on digital twins, enabling closed-loop management of community disasters and timely communication by integrating community-level data, images, and models. Similarly, Cao et al. [17] reviewed the integration of BIM and Geographic Information System (GIS) in urban disaster management, highlighting the potential for real-time data acquisition, interoperability, and data utilisation. Cheng et al. [18] proposed a spatially linked continuous image (SLSI) method, which allows real-time 3D reconstruction of disaster areas using optical data from UAVs, addressing the delays often associated with post-disaster image processing.

While these technologies have shown promise, most studies on bridge disaster management primarily focus on specific phases of the disaster cycle, often utilising bridge information systems, structural health monitoring (SHM) systems, or other inspection applications [19–22]. Most of this research emphasises either enhancing the performance of bridges before a disaster occurs or facilitating rapid rescue and recovery efforts after a disaster. Bridges are load-bearing structures where minor damages can quickly escalate into critical failures under disasters. This highlights the importance of real-time, dynamic data integration to prevent potential collapses. Moreover, research on real-time disaster management has predominantly focused on urban environments, with limited in-depth studies focusing specifically on bridges [10]. This points to a critical gap in the integration of advanced technologies within bridge disaster management, which is necessary to ensure comprehensive coverage across all phases of bridge disaster management—from real-time disaster warning and emergency response to disaster assessment and recovery.

Although BIM and DT technologies are commonly utilised during the design, construction, operation, and maintenance phases of bridges [23–25], some research has explored their role in assessing bridge disaster resilience [26]. For instance, Lin et al. [27] proposed a method for evaluating the collapse susceptibility of large-span cable-stayed bridges under seismic activity using DT models. Their findings demonstrated that the DT approach closely aligns with experimental results, providing accurate predictions of earthquake intensity and collapse probability. These studies suggest that leveraging BIM and DT technologies can enhance the timeliness, coordination, and efficiency of bridge disaster management by unifying technology systems and improving the flow of critical information across all disaster phases.

By examining these gaps, this paper highlights the urgent need for a comprehensive real-time disaster management system for bridges. It reviews the types of disasters that bridges may encounter, summarises the current application status of real-time disaster management across different phases of the disaster cycle, and introduces the RtBDM framework. This framework aims to achieve technology interoperability and information sharing in bridge disaster management, facilitating iterative and adaptive management practices throughout all phases of the disaster cycle to ensure bridges are more resilient and responsive to disasters. The paper is structured as follows: Section 1 provides the introduction, followed by the overall research methodology in Section 2. Section 3 categorises the types of bridge disasters and identifies the need for real-time disaster management. Section 4 summarises the application of real-time disaster management across each disaster phase of bridges. The proposed RtBDM framework is discussed in Section 5, and the conclusion is presented in Section 6.

2. Methodology

In this paper, a systematic review methodology was employed, adhering to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework, to critically evaluate and synthesise existing literature on real-time bridge disaster management. The PRISMA framework provides a structured approach for conducting and reporting systematic reviews, ensuring transparency and reproducibility in the research process [28]. Data were systematically extracted and analysed from publications focusing on disaster types, enabling technologies, and application frameworks. The findings were synthesised to provide a holistic understanding of current advancements and gaps in the field, ultimately informing the development of a conceptual real-time bridge disaster management framework. The process of article retrieval and screening, as illustrated in Fig. 1, further supports the systematic approach taken in this research.

- **Identification**. The process of selecting articles for the systematic review on bridge disaster management began with a comprehensive search using the Web of Science Core Collection and Scopus as the target databases for literature search and analysis. For Web of Science, the search utilised “TS” (Topic) as the main field, encompassing the subject, abstract, and keywords of the paper, while for Scopus, the equivalent field “TITLE-ABS-KEY” was employed. Advanced Boolean operators (“AND” and “OR”) were used to generate various keyword combinations, including “Bridge” and “Disaster”, “Disaster management” and “Real-time”, “Digital Twin” and “Bridge”, “Digital model” and “Bridge”, “Bridge” and “Performance” and “Response”. “Performance” and “Response”, “Bridge” and “Recovery”. A total of 574 papers were retrieved from the Web of Science, supplemented by 338 additional papers from Scopus, resulting in 912 papers initially identified. Of these records, 328 duplicate entries were removed to ensure unique data. 96 entries were excluded because they did not belong to English papers published between 1 January 2000 and 1 September 2024. The remaining 488 records entered the next stage.
- **Screening**. Following PRISMA guidelines, articles underwent a screening process. First, only articles, proceeding papers, and reviews were selected, and 36 records were excluded. Following that, the titles and abstracts of the articles in the search results were manually reviewed to assess relevance, followed by a detailed examination of the full text to determine suitability for inclusion. After screening, 97 articles were retained.
- **Included**. The final phase showed the results of the literature collation. Additionally, citation searching, where references from initially selected articles are used to discover further relevant publications, was employed to expand the pool of reviewed literature, as shown in Fig. 1 (right). The PRISMA process is largely the same, except that database searching is replaced by citation searching in

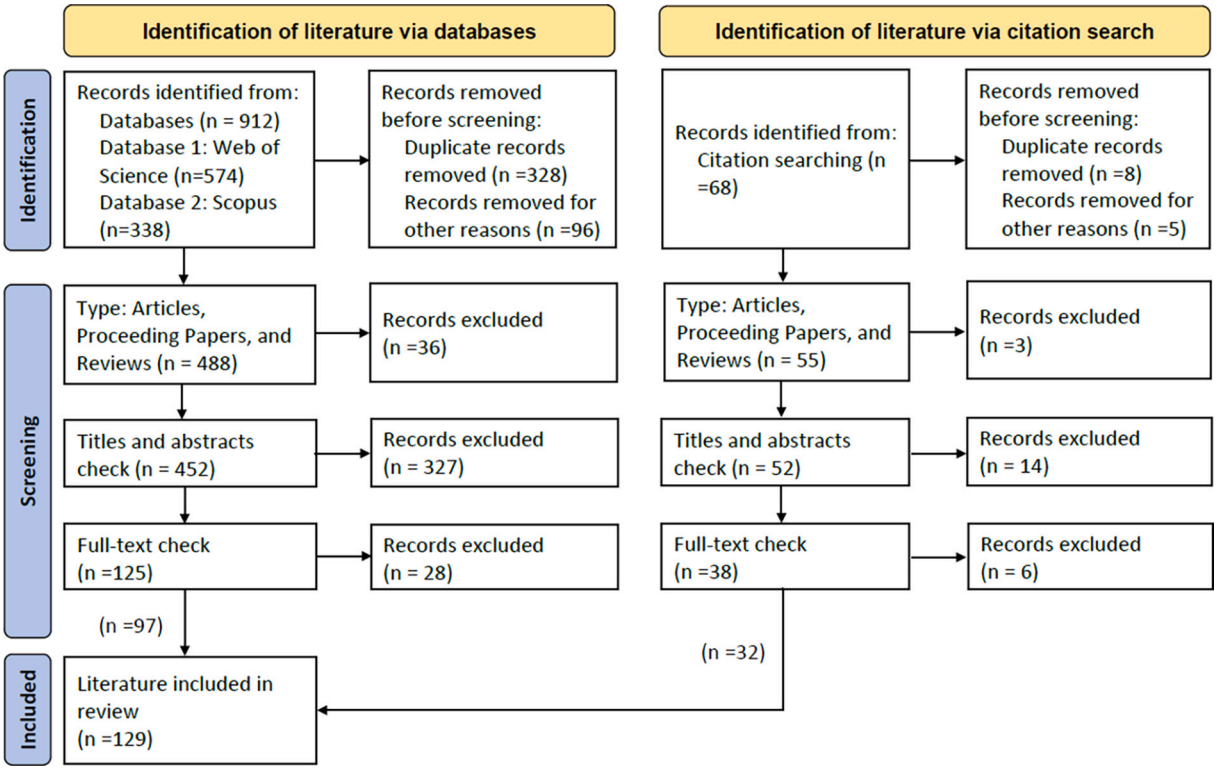


Fig. 1. Process of article retrieval and screening.

the identification phase, and 32 records are finally screened. The final total number of core papers used for the review was 129. Ultimately, all chosen articles were compiled for the final systematic review, providing a comprehensive overview of the field. The information visualisation software CiteSpace was used for the scientometric analysis of keywords and cited journals.

The keyword co-occurrence network and journal co-citation network are illustrated in Fig. 2. Keywords are used to encapsulate authors' academic perspectives, and a keyword co-occurrence analysis is employed to identify research hotspots within the field of bridge disaster management. The node type was designated as "keyword," and CiteSpace was employed to analyse the keywords and generate a keyword visualisation map. In Fig. 2, the node size indicates the frequency of keyword occurrences, while the links between two nodes signify their co-occurrence in at least one article. The analysis reveals that articles in this field primarily focus on framework, disaster management, modelling, BIM, DT, design, and bridge operations and maintenance.

The journal co-citation network is illustrated in Fig. 3 and Table. 1 shows the top 10 high-impact journals between 2000 and 2024: Automation in Construction (33); Engineering Structures (32); Computer-Aided Civil and Infrastructure Engineering (31); Journal of Computing in Civil Engineering (27); Journal of Bridge Engineering (27); Earthquake Spectra (22); Structure and Infrastructure Engineering (21); Advanced Engineering Informatics (19); Structural Control and Health Monitoring (19); Sensors (16).

3. Bridge disaster series review

Bridges are exposed to multiple hazards, with both natural and human-caused disasters increasing significantly in recent decades [29]. Effective disaster management must account for the diverse types of disasters, as the complexity and continuity of disaster damage can

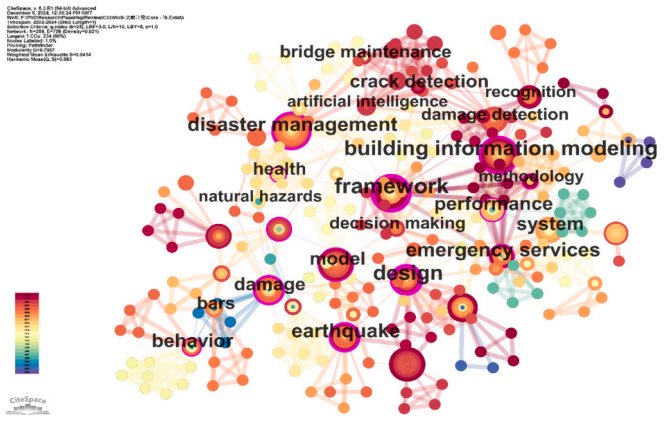


Fig. 2. Keyword co-occurrence network.

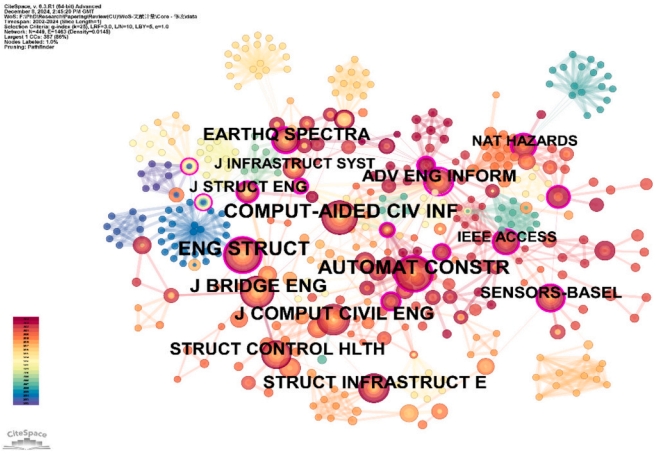


Fig. 3. Journal co-citation network.

Table 1
Top 10 high-impact journals, 2000–2023.

Rank	Journal Title	Co-Citation Frequency
1	*Automation in Construction*	34
2	*Engineering Structures*	33
3	*Computer-Aided Civil and Infrastructure Engineering*	31
4	*Journal of Computing in Civil Engineering*	27
5	*Journal of Bridge Engineering*	27
6	*Earthquake Spectra*	22
7	*Structure and Infrastructure Engineering*	21
8	*Advanced Engineering Informatics*	19
9	*Structural Control and Health Monitoring*	19
10	*Sensors*	16

impact all phases of the disaster response process [30]. Developing efficient bridge disaster management techniques and methods requires a deep understanding of the specific types of disasters, the defects they cause, and the technologies available to address them.

In this paper, bridge-related disasters are classified into three main categories: (1) natural disasters, such as earthquakes, floods, and hurricanes/typhoons; (2) human-caused disasters, including traffic accidents, terrorist attacks, and acts of vandalism; and (3) structural disasters, such as material deterioration, and design defects. Understanding the relationship between these disasters and the defects they cause is critical for selecting appropriate sensing technologies, integrating real-time monitoring systems, and developing effective decision-making strategies.

Real-time disaster management provides a systematic framework for managing disasters and their consequences. It comprises four key phases: Pre-Disaster Warning, Real-Time Disaster Response, Immediate Post-Disaster Response, and Post-Disaster Recovery. Each phase leverages advanced monitoring, data collection, analytical tools, and decision-making technologies to mitigate damage, enable emergency response, and support recovery efforts. The following subsections explore the types of disasters, their associated bridge defects, and the role of real-time disaster management in addressing them.

3.1. Natural disasters

Earthquakes represent one of the most significant natural disasters affecting bridges globally, characterised by their suddenness and destructiveness. In recent decades, frequent large earthquakes have caused severe damage to bridge structures, including tilting or collapse, damage to piers and abutments, foundation settlement or lateral shift, cracks or collapse of decks, and displacement of connecting parts. Among these, substructure damage is one of the most commonly reported issues [31,32].

Seismic forces generated by earthquakes often result in shear or flexural cracks in piers and abutments due to horizontal ground acceleration [33]. Inadequate seismic joints may lead to bridge girder misalignment, while poor foundation stability can cause lateral shifts or settlements [34]. For instance, during the Wenchuan earthquake, 23.5 % of bridges sustained moderate to severe damage or complete failure, with substructure damage being one of the most significant issues (Fig. 4). Monitoring these defects requires technologies such as accelerometers and tilt sensors to detect real-time displacement, as well as strain gauges for measuring stress concentrations at critical points. In real-time disaster management, earthquakes primarily require technologies for the real-time disaster response and immediate post-disaster response phases. During the real-time response phase, accelerometers and GPS systems monitor real-time structural displacement, enabling emergency managers to assess immediate risks and guide evacuation or reinforcement measures. In the immediate post-disaster response phase, tools like finite element modelling (FEM) and digital twins (DT) are employed to evaluate the residual strength of damaged components, simulate structural performance, and prioritise repair strategies.

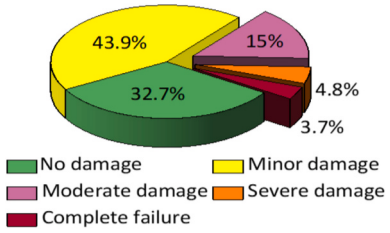


Fig. 4. Proportion of bridges experiencing different levels of damage (no damage, minor, moderate, severe, and complete failure) in the VII–XI intensity zones of the Wenchuan earthquake. The chart highlights the significant vulnerability of bridges to seismic events, with 23.5 % of bridges sustaining moderate to severe damage or complete failure reproduced from [32].

Moreover, earthquakes often trigger cascading hazards, such as landslides, floods, or tsunamis, which exacerbate the initial damage. For example, landslides triggered by the Wenchuan earthquake caused additional foundation instability in many bridges. This highlights the importance of integrating multi-hazard monitoring systems that integrate seismic, geotechnical, and structural data to account for both primary seismic events and secondary hazards.

Rapid urbanisation, global warming, and erratic climate change have dramatically increased flooding, which now accounts for one-third of the total economic losses from natural disasters [35,36]. During floods, bridges experience a progressive sequence of defects, including: (1) foundation scour caused by high-velocity water flows, which is the primary trigger for structural instability, (2) reinforcement corrosion accelerated by prolonged water exposure, (3) abutment erosion from sustained water pressure, and (4) deck damage from debris impact. These defects often occur progressively, with scour and corrosion creating conditions for more severe structural damage [37]. Scour is the leading cause of flood-induced bridge failure, as it undermines the stability of foundations, potentially leading to collapse. For example, Kim et al. [38] evaluated the vulnerability of bridges under floods by analysing failure modes such as reinforcement fracture and abutment instability. Monitoring these defects requires sonar imaging systems to assess scour depth and electrochemical sensors to detect early signs of reinforcement corrosion.

Floods emphasise the importance of the pre-disaster warning and post-disaster recovery phases in real-time disaster management. In the pre-disaster warning phase, real-time monitoring of rainfall intensity, water levels, and scour depth using IoT-based systems and sensor networks can provide predictive insights. For example, advanced multi-source data fusion systems integrate geological, meteorological, and bridge health data to inform managers about potential flood risks and enable proactive measures. During the post-disaster recovery phase, hydrodynamic models and BIM-DT integration are used to simulate foundation damage, assess long-term risks, and optimise rehabilitation plans. Despite advancements, gaps remain in integrating real-time monitoring into comprehensive flood models. Current research lacks systems that effectively combine real-time environmental and structural data for immediate risk assessment and response [39–41].

During hurricanes, bridges are exposed to concurrent hazards from strong winds, waves, and storm surges, posing major threats to the coastal transportation system. Typical damage includes deck misalignment caused by wind uplift forces, displacement of girders under wave impact, and bearing damage caused by simultaneous wind and wave forces [42]. For example, Hurricane Katrina damaged nearly 45 bridges, primarily affecting the superstructure, including shifting or dislodging of main girders, damage to guardrails, and falling bearings [43]. Monitoring these defects requires wind tunnel simulations to predict wind-induced forces and displacement sensors to measure girder shifts.

Hurricanes highlight the importance of real-time disaster management across all four phases. In the pre-disaster phase, predictive analytics using wind tunnel simulations and weather data help anticipate

storm impacts, allowing for proactive measures such as bridge closures or reinforcements. During real-time disaster response, UAVs and drones are rapidly deployed to assess structural conditions and identify critical damages immediately after the storm. In the immediate post-disaster phase, digital twins (DT) and structural analysis models simulate hurricane-induced damage to guide temporary repairs effectively. Finally, in the post-disaster recovery phase, insights from earlier stages are used to implement retrofitting solutions, such as improving wind resistance or reinforcing bearings, to enhance resilience against future hurricanes. Nonetheless, existing research primarily focuses on post-disaster bridge damage assessment using digital technology, lacking disaster management measures for the entire lifecycle of the bridge [44,45]. Additionally, hurricanes also highlight the need for multi-hazard monitoring systems, as storm surges frequently lead to flooding, which causes additional scour and corrosion damage. Comprehensive real-time systems that integrate wind, wave, and structural monitoring technologies are essential for ensuring effective mitigation, response, and recovery.

3.2. Human-caused disasters

Apart from natural disasters, bridges are affected by human-caused disasters such as traffic accidents, terrorist attacks, or fires. Traffic accidents primarily cause damage to the bridge members, induce vibration effects from vehicle impacts, support system damage, and ancillary facilities damage. For example, heavy vehicles colliding with bridge components can lead to localised structural damage, requiring strain sensors and impact detection systems to monitor and assess the effects. In real-time disaster management, traffic accidents are primarily addressed during the real-time disaster response phase, where UAVs, impact sensors, and mobile monitoring tools are deployed to assess immediate damage and guide emergency repairs. Current real-time monitoring systems for traffic-related damage remain underdeveloped, with most assessments relying on post-accident inspections.

Transportation infrastructure is also a common target of terrorist attacks; for instance, passenger trips decreased by 6 % in the two years following the 9/11 attack [46]. Terrorist attacks involving explosives often result in structural cracks, pier fractures, or loss of load-bearing capacity. Monitoring for terrorist-related damage requires blast sensors and structural health monitoring (SHM) systems integrated with emergency response protocols. Terrorist attacks emphasise the need for real-time disaster response and post-disaster recovery phases. SHM systems provide critical data for assessing structural safety immediately after an attack, while BIM-DT integration supports long-term recovery planning. The ongoing process of urbanisation and industrialisation has resulted in the construction of more bridges, increasing the risk of such disasters.

Bridge fires are mainly caused by car accidents and gasoline combustion rather than disasters such as earthquakes or ship collisions [47,48]. Additionally, the increased transportation of flammable materials has led to a subsequent increase in the frequency and severity of fire accidents, potentially leading to bridge collapse. The effects of fire-induced damage include loss of concrete strength, reinforcement degradation, and deformation of structural components, all of which significantly reduce load-bearing capacities [49]. For instance, the I-85 bridge fire in Atlanta caused severe structural damage and prompted significant transportation disruptions. Common post-fire assessment methods include visual inspection and material strength testing, with various manuals and guidelines available for post-fire assessment and rehabilitation strategies [50–52]. However, real-time fire monitoring systems remain underdeveloped, with no standards for developing pre-fire prevention and emergency response plans for bridge fires. Fires highlight gaps in pre-disaster prevention and real-time response phases. Addressing these gaps is important for minimising fire-related damage and ensuring timely recovery. Current monitoring, warning, and assessment of bridges mainly focus on natural disasters and structural deterioration, leaving gaps in effective countermeasures for traffic accidents and relative damage. Discrepancies in standards and codes across different regions further complicate efforts to harmonise practical solutions.

3.3. Structural disasters

Structural disasters are mainly caused by design defects and material deterioration, both of which lead to reduced mechanical properties and disaster resistance [53]. Design defects can stem from irrational design, poor construction quality, or substandard material. For instance, insufficient reinforcement ratio or suboptimal concrete quality can lead to early-stage cracking under operational loads. Moreover, improper pier alignment during construction may result in long-term structural deformation [54]. During construction, structural imbalances can cause accidents, while in the operation and maintenance phase, bridges can suffer from cracks, deformation, and corrosion, potentially leading to partial or total collapse. Addressing damage caused by bridge structural design defects requires a comprehensive approach. BIM-based bridge management systems (BMS) provide decision support throughout the bridge's lifecycle [12]. BIM technology facilitates collaborative design, conducts thorough structural safety inspections, simulates construction processes, and formulates response measures. Additionally, when combined with advanced inspection technologies, BIM enhances the evaluation and analysis of the bridge performance, thereby mitigating disasters caused by structural design defects and material deterioration [55,56].

Material deterioration manifests as a gradual decrease in bridge strength and stability due to factors like reinforcement corrosion or concrete ageing. Corrosion reduces the reinforcing steel's area, weakens the bond strength between concrete and steel, and causes concrete spalling. These issues are often compounded in marine environments, where saltwater accelerates the deterioration process, resulting in progressive structural instability. This, in turn, exposes more steel to external environments, accelerating the deterioration process. If not maintained promptly, structural damage accumulates and intensifies, increasing vulnerability to extreme loading conditions [57,58]. Addressing bridge corrosion involves both prevention and reparative measures, including using corrosion-resistant materials and coatings during design and construction and repairing corrosion-damaged areas through coating repairs and reinforcement. Real-time monitoring systems, such as SHM integrated with DT models, can provide continuous updates on structural integrity, enabling early detection of material deterioration and preventing catastrophic failures. Table 2 illustrates the relationship between different disaster types and associated bridge damage. Understanding these disaster-defect relationships is critical for selecting appropriate sensing technologies and decision-making strategies.

3.4. Need for real-time disaster management

Disaster management involves managing the risks and consequences of disasters, typically divided into four phases: mitigation, preparedness, response, and recovery. This process includes activities such as risk assessment, pre-disaster preparedness, emergency response, rescue operations, and post-disaster reconstruction [59,60]. Callaghan [61] highlights that the field of disaster management is a crucial area of development for achieving real-time capabilities, emphasising the importance of "real-time" in saving lives and reducing losses. In the context of bridges, real-time disaster management is particularly important for ensuring life safety, preserving bridge structures, and maintaining their functional use. This approach leverages advanced sensor technologies, communication networks, data analysis and processing systems, and automated decision-making mechanisms to enable rapid and effective responses during disasters.

According to the definition of disaster management, the disaster

Table 2
Disaster types, typical defects, and potential impacts for bridge structures.

Disaster type	Typical defects	Potential impacts	Monitoring Technologies
Earthquake	Deck misalignment, pier deformation, foundation settlement or lateral shift, and connecting parts displacement.	Structural instability, reduced load-bearing capacity, and increased vulnerability to secondary disasters.	Accelerometers, GPS, tilt sensors
Flood	Scour, reinforcement corrosion, foundation erosion, and bridge deck damage.	Foundation collapse, reduced structural strength, and loss of connectivity.	Sonar imaging, electrochemical sensors
Hurricane	Shifting or dislodging of girders, damage to guardrails, and falling bearings.	Service disruption and partial collapse of superstructure.	Wind tunnel simulations, displacement sensors
Traffic accidents	Vibration effects, support system damage, and damage to ancillary facilities.	Localised structural damage and potential service interruptions.	Impact detection sensors
Terrorist attacks	Structural cracks, pier fractures, and loss of load-bearing capacity.	Structural damage and service disruption.	Blast sensors, SHM
Fire	Concrete spalling, concrete strength reduction, and reinforcement degradation.	Reduced structural strength, partial or total collapse.	Thermal sensors, material testing
Design Defects	Structural imbalances, inadequate safety margins, cracks or deformation.	Reduced disaster resistance, accelerated ageing of materials, partial or total collapse.	BIM, SHM
Material Deterioration	Reinforcement corrosion or concrete ageing.	Gradual loss of strength and increased vulnerability to extreme loading conditions.	Corrosion sensors, strain gauges

process under real-time disaster management is divided into four phases: Pre-Disaster Warning, Real-Time Disaster Response, Immediate Post-Disaster Response, and Post-Disaster Recovery, as shown in Fig. 5. The pre-disaster warning phase consists of mitigation and preparedness. Mitigation refers to preventing a disaster or reducing the impacts should one occur. Preparedness is in terms of pre-disaster planning, training, and preparation. In the pre-disaster warning stage, real-time disaster management leverages advanced sensor networks and multi-source data fusion technology to integrate geological, meteorological, and bridge health monitoring data. This integration enables the real-time warning system to provide accurate predictive information before a disaster occurs, aiding managers in taking preventive measures [62]. For real-time disaster response, the management emphasises the rapid acquisition and processing of disaster data to support real-time decision-making and emergency actions. Technologies such as drones, robots, and mobile monitoring equipment are employed to swiftly gather real-time data from disaster sites. Combined with BIM and DT technologies, this data facilitates dynamic analysis and evaluation of bridge structures, providing emergency decision support and temporary reinforcement solutions [63].

In the immediate post-disaster response phase, real-time disaster management focuses on detailed damage assessment and preliminary repair of bridges. Utilising damage detection technology and data analysis tools, it quickly assesses the extent of bridge damage and performance status, subsequently developing preliminary repair plans [22]. During the post-disaster recovery phase, real-time disaster management emphasises long-term rehabilitation and functional recovery of bridges. Through virtual simulation and optimisation, it formulates a more scientific and efficient long-term rehabilitation plan, ensuring that bridge performance levels are restored to pre-disaster conditions or higher [64].

Overall, real-time disaster management enhances the resilience and efficiency of bridges by implementing comprehensive and dynamic monitoring and management at all stages of a disaster. However, according to existing research, further advancements are needed in technical tools related to data integration, real-time analysis, and decision support to achieve more efficient life cycle management. Existing real-time systems face challenges such as data integration from diverse sources, latency in decision-making, and the high costs of implementing advanced monitoring technologies, which must be addressed to ensure comprehensive disaster management solutions.

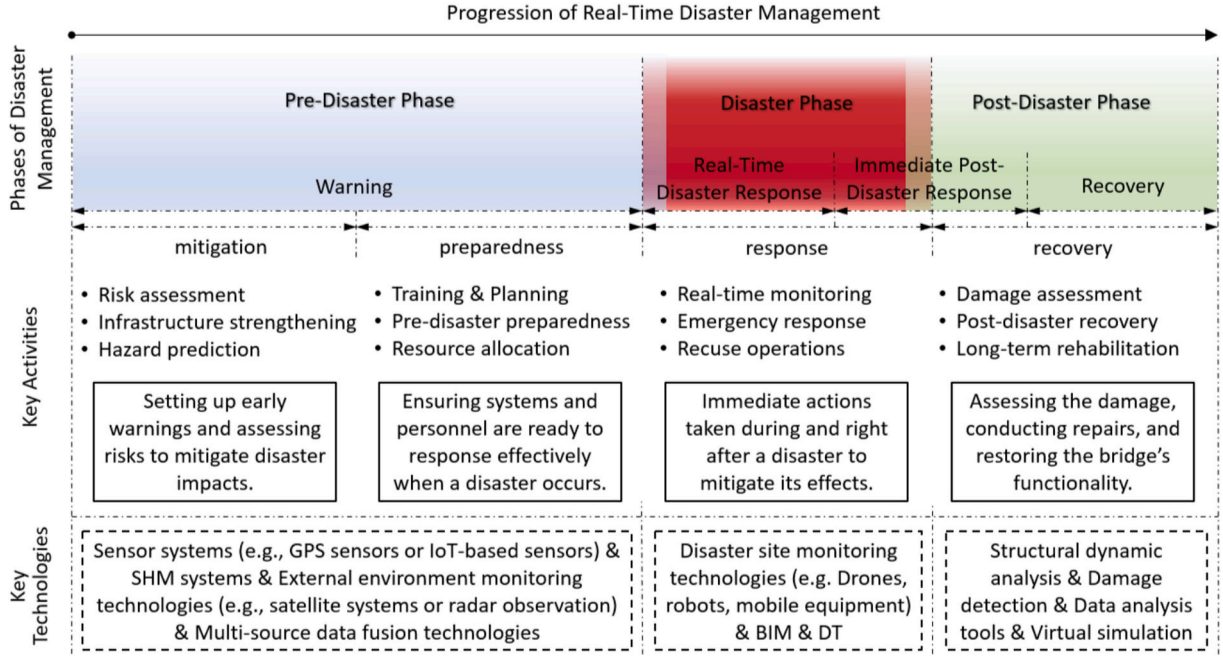


Fig. 5. Phases, key activities, and key technologies in real-time disaster management.

4. Technological review for bridge disaster management

The preceding discussion in Section 3 categorised disasters and their corresponding bridge damage, providing a foundational understanding of the scenarios in which specific technologies are applied. This section expands on those insights by aligning the disaster management phases with appropriate technological responses. Real-time disaster management for bridges involves a systematic approach that spans all phases of the disaster management cycle, from pre-disaster warning to post-disaster recovery. This section is divided into four phases: Pre-Disaster Warning, focusing on early risk monitoring and predictive analytics; Real-Time Disaster Response, which ensures rapid data acquisition and emergency decision-making; Immediate Post-Disaster Response, emphasising damage assessment and initial repair strategies; and Post-Disaster Recovery, targeting rehabilitation and performance restorations for bridges.

It should be noted that while disaster management encompasses four phases, not all scenarios fit neatly into all cycles. For instance, sudden-onset disasters like earthquakes often lack a significant pre-disaster warning phase, whereas long-term recovery tends to dominate in flood scenarios. By exploring the application of these advanced technologies, this section highlights their advantages in enhancing disaster management efficiency, evaluating how these technologies and strategies align with the specific phases relevant to each disaster while acknowledging existing limitations, such as data integration challenges and gaps in real-time analysis capabilities.

4.1. Pre-disaster warning

Pre-disaster warning is the initial phase of the disaster management cycle in real-time disaster management, as illustrated in Fig. 5. The phase involves the real-time collection and fusion of monitoring data from bridge structures and the surrounding environment, triggering alarms when abnormal signs or structural states are detected [65]. For this phase, bridge-specific defects such as scour, corrosion, and structural deformation caused by slow-developing hazards or predictable events (e.g., floods, hurricanes) are the primary targets for monitoring. However, sudden-onset disasters like earthquakes, fires, and traffic accidents often lack an effective pre-disaster warning phase due to their unpredictability.

Disaster warning consists of four components: (1) risk knowledge, (2) monitoring, (3) communication of warnings, and (4) response capability. Risk knowledge involves the systematic collection and analysis of historical and real-time data to understand potential hazards and vulnerabilities, forming the foundation for informed decision-making. Monitoring refers to the dynamic process of collecting real-time data and analysing it in the context of existing knowledge to predict potential risks and trigger timely interventions. Communication of warnings focuses on the effective dissemination of data, information, and insights related to early-warning situations, ensuring that relevant stakeholders are informed promptly and accurately. Finally, response capability emphasises the importance of preparedness and the ability to take swift, decisive action based on warning signals to mitigate potential impacts. These components interact synergistically to form an effective warning system [66,67]. The components are defined as shown in Table 3.

Table 3
Elements of disaster early warning adopted from [68].

Component	Definitions of Component
Risk knowledge	Data collection and analysis of risks.
Monitoring	Ability to collect dynamic data and analyse it to make decisions based on existing knowledge.
Communication of warnings	Sharing and transferring data, information, and knowledge on early-warning situations.
Response capability	Preparedness to know how to take action.

Monitoring and predictive analytics are crucial for the pre-disaster warning phase, particularly for disasters such as earthquakes, floods, and hurricanes, which have varying levels of predictability. For example, for flood-induced defects, such as scour and reinforcement corrosion, are monitored using sonar imaging and electrochemical sensors, while machine learning (ML) models predict potential scour based on rainfall and water flow data [69]. However, disasters such as earthquakes, fires, traffic accidents, and terrorist attacks generally lack a distinct pre-disaster warning phase due to their sudden nature. Similarly, material degradation represents a slow-developing hazard, and early warning focuses on long-term monitoring rather than immediate disaster prevention.

Advanced sensor networks and data fusion technologies are extensively employed in bridge pre-disaster warnings. Sensor networks collect real-time data, including vibration, displacement, and strain, by deploying various sensors at critical points on bridges and transmitting this data through wireless networks for analysis [70]. For example, Jo et al. [71] demonstrated high-precision bridge displacement assessments using relatively low-precision GPS sensors, while Hou et al. [72] designed an IoT-based sensor system for bridge displacement monitoring utilising NB-IoT technology. This system enables real-time data transmission, analysis, and remote visualisation through a web interface. While IoT-based systems enhance data collection and transmission, challenges remain in ensuring data compatibility across heterogeneous sensor networks and minimising transmission delays in disaster scenarios [1,73]. These limitations highlight the need for robust data processing protocols and reliable communication networks to ensure continuous monitoring.

SHM system provides real-time data through sensors and visual inspection, while BIM simulations of the bridge's internal structure offer condition feedback and timely warnings before disasters [74,75]. Jeong et al. [76], for instance, integrated a BIM-based bridge information model (BrIM) with SHM to enhance bridge monitoring by utilising the OpenBrIM standard for data reception, storage, and analysis. However, the lack of standardised protocols for data exchange between BIM and SHM systems often results in integration inefficiencies. Moreover, BIM's reliance on accurate historical data makes it less adaptable to sudden and dynamic disaster scenarios [19].

With the continuous inflow of data, it is essential to delve into the vast amounts of collected quantifiable information from the data layer. Data-driven analytics can significantly enhance the assessment of structural, geographical, and environmental conditions [77]. For example, predictive models based on ML and deep learning can identify potential disaster risks through real-time data analysis. Deng et al. [78] proposed a technique that automatically identifies concrete cracks using computer vision and machine learning, triggering pre-disaster warnings to prevent disasters due to structural damage.

External environment monitoring primarily utilises technologies such as wireless sensor networks (WSN), synthetic aperture radar interferometry (InSAR), Light Detection and Ranging (LiDAR), and satellite systems like GPS and Beidou. Lee et al. integrated [70] a WSN and smart building technologies to develop an IoT-based bridge safety inspection system for monitoring and analysing bridge and environmental conditions in real-time. Zhang et al. [79] developed a local wind environment monitoring system for large-span bridges in mountainous areas, enabling accurate assessment of regional wind field parameters. These parameters provide a critical reference for the wind-resistant design of bridges. Similarly, Frigerio et al. [80] developed a web-based platform for continuous landslide monitoring that is capable of detecting displacements and triggering disaster alerts. Liu et al. [81] established a tsunami prediction system using global satellite positioning technology.

With advances in wireless sensor networks, cloud computing, communications, and other fundamental technologies, DT technology has been initially applied in environmental monitoring [20]. For example,

Dan et al. [20] established a DT system for a regional transportation infrastructure network that uses measured traffic loads as a link to monitor working conditions and issue safety warnings for all bridges in the network. Kang et al. [23] proposed a DT-based bridge health monitoring method, synchronising physical and virtual spaces using sensor data to update digital models and simulate extreme conditions, ensuring bridge health. Despite advancements in monitoring, the implementation of digital Twins (DT) often demands substantial computational resources and dependable sensor data inputs, which may become constrained during large-scale disasters. Integrating DT with complementary technologies, such as AI and edge computing, enhances its adaptability and efficiency in pre-disaster warnings [82].

Early warning indicators and thresholds, based on historical data, norms, and theoretical analyses, are crucial. They help determine responses by comparing real-time monitoring data with established thresholds. These indicators must be sensitive to structural abnormal changes, and the appropriateness of warning thresholds directly affects the system's ability to detect and signal issues [83]. For instance, Lattanzi et al. [84] verified the strong correlation between the degree of damage to reinforced concrete piers and the pier top displacement by combining image segmentation and feature extraction with nonlinear regression analysis. They explored the use of computer vision and machine learning for quantitative assessment of bridge conditions. Similarly, Yan et al. [85] employed drones with cameras and LiDAR scanners to automatically detect and quantify concrete cracks for structural assessment. The diversity of warning indicators results in various warning methods, which may cause public misunderstanding. Neußner [86] reviewed approximately 110 warning methods used in different countries and regions, suggesting that standardised warning signals, including colours, numbers, symbols, words, or simple messages, should be established.

The insights gained from pre-disaster warnings, such as real-time monitoring data and predictive analytics, are foundational for informing rapid decision-making during the real-time disaster response phase, as discussed in Section 4.2. However, while current early warning technologies are effective in predicting and warning of potential disaster risks, limitations in data transmission and real-time analysis persist. Although BIM and DT technologies have improved pre-disaster warning accuracy and response speed by integrating historical and real-time data for dynamic simulation and prediction, a systematic real-time management framework has not yet been developed [87]. With the increasing demand for accurate disaster prediction, it is crucial to mine the data to provide users with timely feedback for prediction and decision-making. Therefore, more attention should be paid to integrating BIM and DT with technologies like IoT and data mining to enhance real-time early warning, enabling closed-loop management and dynamic updating of pre-disaster warning information [88,89].

4.2. Real-time disaster response

The survival probability in a disaster area decreases significantly after 72 h, making this period critical for disaster response [90,91]. The real-time disaster response phase, defined as the 72 h following a disaster, forms a crucial component of real-time disaster management. It should be clarified that real-time disaster management for bridges emphasises the real-time management of the whole life cycle, focusing on ongoing monitoring and early warning during the operation and maintenance phases, as well as providing integrated emergency response and post-disaster recovery management. The goal is to ensure that managers can take action based on the most up-to-date analyses at any point in time. In contrast, the real-time nature of disaster response specifically addresses the immediate aftermath of an emergency event.

Bridge damage can severely disrupt transportation routes vital for rescue operations, medical evacuations, and supply chains. This underscores the critical need for rapid and accurate bridge damage assessment, utilising methods specifically designed to address the dynamic and load-bearing characteristics of bridges under disaster conditions [21]. Historically, post-disaster bridge inspections relied heavily on manual investigations, which, although precise, are often time-intensive and costly. These traditional approaches fail to meet the urgent needs of real-time disaster response [92]. It should be noted that, unlike these sudden-onset disasters, material degradation lacks a real-time response phase, and its slow progression requires continuous monitoring rather than immediate. Modern technologies, such as satellites, drones, and rescue robots, have transformed this phase by generating vast amounts of real-time disaster-related data, significantly improving the speed and efficiency of assessments [93]. For instance, remote sensing technology combines sensors from satellites and ground-based platforms to establish a three-dimensional remote sensing network capable of acquiring real-time images to analyse the type and extent of structural damage [94]. In earthquake scenarios, high-frequency accelerometers and GPS systems monitor real-time pier displacement and girder oscillations [95]. For flood events, sonar imaging and LiDAR are employed to evaluate underwater scour and foundation erosion. These methods are particularly effective for large-scale regional assessments but may lack the resolution required to identify fine-scale structural damage in critical bridge components.

Several advanced technologies have been proposed to improve real-time disaster response tasks. For instance, Chen et al. [92] leveraged high-resolution optical remote sensing imagery to detect damage in water bridges, enhancing the timeliness and accuracy of post-disaster assessments. Similarly, Congress et al. [96] introduced a UAV-CRP (Unmanned Aerial Vehicle Combined with Close Range Photogrammetry) technique for rapid, efficient, and repeatable evaluations of bridge damage. UAV-CRP excels in capturing detailed imagery of surface-level damage but is limited in assessing internal structural conditions, such as force reduction or material degradation. To address this limitation, Zhao et al. [97] developed a smartphone-based cable force measurement system that combines GPS data, structural images, and portable sensing capabilities, providing critical information for decision-making during real-time disaster responses. While these tools enable rapid data acquisition and initial damage assessments, they are limited by their reliance on discrete data acquisition and static analysis, which fail to capture dynamic changes in bridge conditions during a disaster.

The constraints of disaster environments necessitate the use of portable and easy-to-operate equipment by emergency rescue personnel. UAVs equipped with computer vision techniques have proven effective for real-time bridge inspections. Khallo et al. [98] used Unmanned Aerial Vehicle (UAV) and computer vision techniques to integrate a combination of UAV design, data capture, and data analysis to speed up bridge inspection without impeding access. However, the UAV-based method typically relies on discrete data acquisition and requires manual interpretation, which can delay the integration of findings into a comprehensive disaster response plan [99]. Bridges, serving as lifeline routes for post-disaster rescue and relief, are critical for transporting search and rescue personnel and medical teams and evacuating the injured from disaster areas. Chikahiro et al. [100] developed a new type of emergency mobile bridge, where the components are combined into an X-shape by a pivoted pin joint, allowing for rapid transport and the construction of a bridge.

These techniques facilitate the rapid acquisition of bridge damage data and conditions after a disaster, providing real-time images and structural status information. Data collected in these disaster scenarios, such as real-time displacement, vibration frequencies, and water flow velocities, which feed into Finite Element Models (FEM) for structural diagnostics and prognostics. However, they are limited in their ability to perceive dynamic changes in bridge status during a disaster. For instance, although UAV and smartphone-based techniques effectively facilitate immediate post-disaster assessments, they lack real-time feedback mechanisms to monitor evolving conditions, such as progressive structural degradation caused by aftershocks or sustained loads

[101]. Most of these techniques rely on discrete data acquisition and static analysis, lacking the continuous monitoring and real-time response needed to fully capture the dynamic changes in bridges, hindering comprehensive and real-time emergency response [102]. Despite these limitations, these tools contribute to the study's objective of improving disaster management efficiency by enabling rapid data acquisition and initial response, which are critical elements of the real-time disaster management cycle.

Recent advancements in real-time disaster response have notably improved the rapid assessment of bridge damage and data transmission. However, these efforts are predominantly confined to the emergency phases following a disaster, focusing on the urgent requirements to assess and implement emergency measures quickly [94]. This highlights a critical gap in system integration, where real-time data collected during pre-disaster monitoring and disaster response is not effectively linked to inform recovery strategies. Considering the four phases of real-time bridge disaster management, current research reveals significant limitations in system integration and information sharing across these stages. For instance, the weak linkage between pre-disaster warning data and real-time disaster response limits the ability to make informed decisions during emergency responses, resulting in a lack of support from preliminary data. Additionally, feedback and utilisation of real-time data are less considered in the post-disaster recovery phase, leading to a disconnect between long-term recovery strategies and immediate disaster response.

DT technology-driven emergency response plans can infer the evolutionary trend of emergencies through knowledge retrieval based on the potential risks perceived in the pre-disaster warning phase [21]. These plans, based on the location of emergency resources, personnel deployment, and potentially hazardous areas, are visually presented for decision-making [103]. As disaster events unfold, resource information may become obsolete when shared. Utilising disaster information systems shared by various agencies and organisations, such as the Disaster Information Integration and Sharing System (DISIS), GIS, and Shared Information Management Platforms (SIMPs), can address this challenge, as shown in Table 4. These disaster information systems are designed for distinct purposes yet collectively address the challenges of resource obsolescence and dynamic information needs during disasters. By enabling data integration, real-time analysis, and effective communication across agencies, these systems facilitate improved coordination and decision-making throughout the disaster management lifecycle. Based on this, BIM can serve as a starting point for DT, enriched by real-time data to provide accurate decision support for real-time bridge disaster response.

In conclusion, real-time disaster response necessitates the timely acquisition, integration, and analysis of data from multiple sources, including satellites, UAVs, sensors, and social media. While recent advancements have improved the speed and accuracy of assessments, challenges in system integration, data transmission, and real-time feedback mechanisms persist [106,107]. Future research should focus on developing intelligent management systems that integrate real-time data from diverse sources, enabling rapid execution of rescue and relief programs. Enhancing the interoperability of technologies like BIM, DT, and disaster information systems will further strengthen the real-time response phase and its contribution to overall disaster management.

Table 4
Different disaster information systems.

Category	Representative System	Purpose
DISIS	Disaster Operations, Risk, Abatement, and Management System (DIORAMA) [104]	Support activities necessary before, during and after a disaster through data access, spatial queries and information sharing across multiple users.
GIS	Web-based Geographic Information System (WebGIS) [16]	Analysing, sharing and visualising geographic information for real-time disaster assessment and emergency response.
SIMPs	Intelligent Process-aware Information Systems (IPAISs) [105]	Support agility in disaster response operations by providing information and process scoping to manage dynamic interoperability.

4.3. Immediate post-disaster response

The National Cooperative Highway Research Program (NCHRP) report [108] defines the period from day one to day seven after a disaster as the post-disaster response recovery period. Based on the definition of the real-time disaster response phase as the 72 h following a disaster, this paper defines the period from day four to day seven as the immediate post-disaster response phase. Unlike the real-time response phase, which focuses on immediate structural safety, during the post-disaster response phase, the main tasks include a comprehensive damage assessment of the bridges, temporary transportation services, and preliminary repair work to prevent secondary disasters. The real-time disaster response phase emphasises immediate response and preliminary structural safety assessment, necessitating a less comprehensive evaluation than the immediate post-disaster response phase. In contrast, the immediate post-disaster response phase focuses on detailed damage assessment and preliminary repair, requiring more extensive disaster data. This distinction between phases allows for a more systematic approach to addressing bridge damage, accounting for both immediate needs and long-term recovery.

The introduction of advanced techniques and equipment has enabled detailed damage assessment of bridges after disasters. Various measurement techniques, such as total station (TS), a differential global positioning system (dGPS), and photogrammetry, have been developed to rapidly acquire structural geometry and material property parameters [109–111]. TS and dGPS are suitable for small-scale and high-precision measurements, such as identifying slight deformations in critical bridge components [112]. In hurricane scenarios, infrared thermal imaging is used to detect hidden cracks in bridge decks caused by extreme wind loads, while FEM simulations evaluate residual structural strength to prioritise repairs [113]. In flood-induced scouring cases, ground-penetrating radar, water-penetrating radar, sonar and some specific laser systems can assess the extent of foundation erosion, ensuring targeted stabilisation measures [114]. However, these techniques still require substantial manual operations and post-disaster data processing, leading to incomplete assessment outcomes.

Computer vision technology, which extracts information from digital images, videos, and other media combined with AI and ML techniques, can support post-disaster damage assessment of bridges more quickly, comprehensively, and intelligently [115]. For example, Torok et al. [116] developed a system centred on image-based 3D reconstruction and a new 3D crack detection algorithm for recognising and colour-coding geometric features of cracks from collected post-disaster structural images. The superior performance of convolutional neural networks (CNNs) in image classification and object recognition has led to a widespread application in computer vision [22], [117]. Liang [22] proposed an image-based three-level post-disaster assessment method for reinforced concrete (RC) bridges by combining deep learning for system level, component level, and component local level, significantly improving the assessment accuracy compared to the traditional computer vision model. Li et al. [118] constructed a full convolutional network (FCN) method for detecting multiple damages in structures, which accurately detects four types of damage: cracks, spalling, efflorescence, and holes.

When making response decisions about the overall safety of a bridge after a disaster, contextual information about the design, connectivity, and function of the damaged members in the overall structural system is required, which cannot be provided by computer vision-based damage assessment. BIM stores geometric and semantic data of the structure,

and DT can address the limitations of integrating visual information with BIM [119]. By utilising DT, the integration of BIM with visual inspection and structural analysis models enables a comprehensive damage assessment and prediction of bridge structures, as shown in Fig. 6. This integrated framework for bridge damage assessment combines BIM and DT technologies to enhance evaluation accuracy. The process begins with a disaster site survey using UAVs, IoT, and laser scanners to capture bridge damage characteristics, such as deflection, crack propagation, spalling, and rebar exposure. The collected data are incorporated into a continuously updated digital model, enabling structural analysis to assess key indicators and triggering additional surveys when necessary. This iterative process ensures real-time updates, dynamic condition monitoring, and adaptive management strategies tailored to bridge-specific needs, ultimately improving resilience to disasters. Levine et al. [120] proposed a structural post-earthquake DT framework based on a BIM platform, combining drone imagery, 3D point cloud monitoring, and damage assessment for rapid identification of post-earthquake structural safety. Lin et al. [27] proposed a collapse assessment method based on DT, demonstrating its feasibility and accuracy in assessing the seismic performance of large-span cable-stayed bridges through shaking table tests and structural analysis models.

While BIM and DT enhance the bridge's post-disaster damage assessment capability, integration with the structural analysis model remains inadequate. Structural analysis models are crucial for assessing the structural response and damage level of bridges. However, the current system lacks effective interfaces and integration mechanisms between these models and BIM/DT, which limits the depth of structural mechanics analysis in structural damage assessment. Despite being supported by rich visuals and data, this gap affects the comprehensiveness and accuracy of the assessment results.

To reopen damaged bridges promptly after a disaster, fast and effective repair measures are needed for bridges with varying levels of damage, especially those on critical routes. Karlaftis et al. [121] proposed a method for allocating funds for bridge rehabilitation, where the first stage involves contingency funds to rehabilitate as many bridges as possible to meet minimum operating levels. Bridge piers, as the main load-bearing components, are prone to damage during disasters, and their large residual deformation increases the difficulty and time for repair. New materials, such as ultra-high-performance concrete (UHPC) and carbon fibre-reinforced plastic (CFRP), have shown potential for rapid bridge structure repair. Farzad et al. [122] verified the effectiveness of UHPC in restoring the stiffness, strength, and deformation of damaged abutments through static tests. He et al. [123] utilised externally bonded CFRP to repair RC bridge piers with varying levels of damage, comparing the performance of repaired piers with that of undamaged piers. The results indicated that externally bonded CFRP can quickly repair damaged RC bridge piers and effectively improve stress performance. Accelerated bridge construction, which aims to minimise on-site construction time for RC bridges, is now being applied to quickly repair earthquake-damaged bridges [124,125].

4.4. Post-disaster recovery

Similar to the NCHRP report [108], the Federal Emergency Management Agency (FEMA) stipulates [64] that short-term recovery is the post-disaster emergency relief phase and is measured in days. Medium-term recovery is measured in weeks or months, while long-term recovery may extend over months or years, encompassing the entire reconstruction and recovery period. Post-disaster recovery encompasses both short-term and long-term activities. Short-term rehabilitation aims to restore bridges to minimum operational standards, while long-term rehabilitation strives to return them to their pre-disaster functional status [126]. Accordingly, in real-time disaster management, the post-disaster recovery duration is defined as the period extending from one week after the disaster to several months or even years.

Karamlou and Bocchini [127] calculated the probability of repair time for each bridge component based on specific repair measures and optimised to obtain the best repair time. Tapia and Padgett [128] employed a multi-objective genetic algorithm to optimise post-disaster retrofitting and rehabilitation strategies for bridges, ensuring structural safety while minimising the society, environment, and economic impacts. Minnie and Moon [129] proposed a short-term rehabilitation goal of restoring 60 % of a bridge's functionality, allowing limited truck and freight traffic, with a long-term goal of fully restoring bridge operations.

In the post-disaster recovery phase, the focus is not only on repairing individual bridges but also on ensuring the smooth and coordinated functioning of the entire transportation network [130]. Efficient rehabilitation hinges on optimising the scheduling of bridge tasks, thereby supporting post-disaster recovery management decisions. Bocchini and Frangopol [131] considered bridge type, damage extent, and discount factor to calculate rehabilitation costs and established a bi-objective optimisation model to maximise recovery capacity while minimising the total recovery cost. This approach is most effective for highly prioritised networks with limited resources but may fall short in scenarios involving interdependent recovery tasks across multiple regions. Zhang et al. [132] analysed the interactions between different bridge rehabilitation activities in their recovery scheduling model, demonstrating that

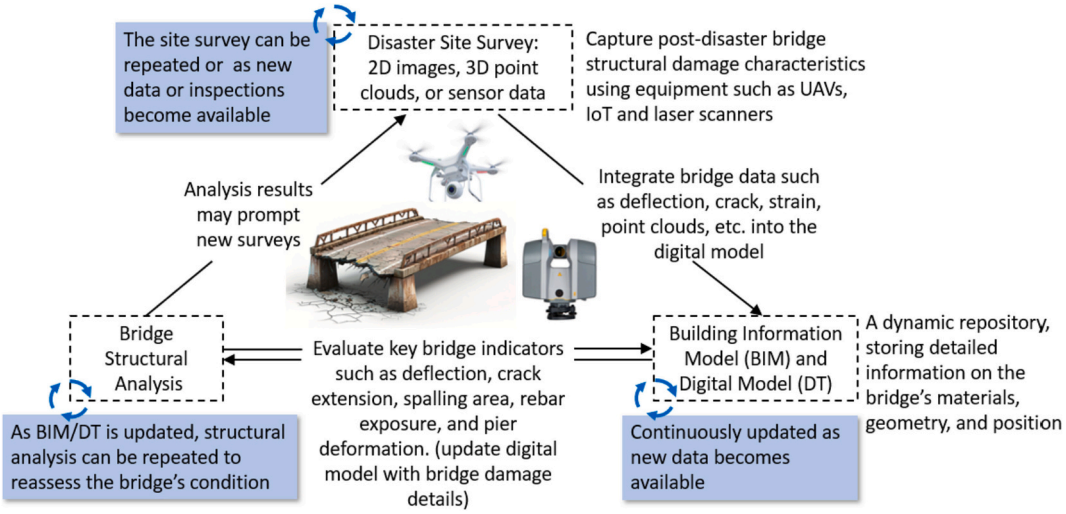


Fig. 6. Bridge damage assessment via integrated BIM and DT.

traditional models, which do not consider these interactions, overestimate the recovery time of bridges after a disaster. Merschman et al. [130] developed performance assessment metrics based on bridge functional, topological, and social indicators to determine the optimal sequence of bridge rehabilitation to maximise road network performance during the recovery period.

Ideally, post-disaster inspection and rehabilitation of bridges should be conducted simultaneously to maximise rehabilitation efficiency. However, these processes interact such that rehabilitation can only commence after inspection, and inspection is typically not feasible for bridges undergoing rehabilitation. To address the challenges, Zhang and Wei [133] modelled the interaction between inspection routes and rehabilitation schedules to determine the optimal sequence for inspecting and repairing damaged bridges.

While current rehabilitation techniques and scheduling models facilitate the development of scientific post-disaster bridge repair measures, the coordination and management of post-disaster recovery remain complex. For instance, bridges affected by earthquakes may require reinforcement of displaced piers, informed by seismic data collected during real-time monitoring. Flood-induced disasters may necessitate the replacement of damaged foundations, guided by pre-disaster hydrodynamic models. Significant challenges persist in information integration and sharing, and short-term decisions made during the disaster response phase can impact the efficiency of long-term recovery efforts [134]. BIM contains component-level information, and DT allows real-time synchronisation between physical and virtual worlds, offering effective scheduling and management. For instance, Sun and Liu [135] constructed a hybrid DT-BIM model that identifies resource shortages, analyses demand, supports decision-making, schedules resources, and updates all processes in the database with the aid of artificial intelligence. Ford and Wolf [15] highlighted that using DT to build a post-disaster information platform enables information sharing among departments and coordination of the recovery process. Xu et al. [136] proposed a building earthquake damage assessment method based on BIM and FEMA P-58. The earthquake damage prediction results from FEMA P-58 are imported into BIM through model mapping and interactive information transfer, allowing for the estimation of earthquake damage and restoration costs.

In summary, the four phases of real-time disaster management—pre-disaster warning, real-time disaster response, immediate post-disaster response, and post-disaster recovery—represent a systematic approach to mitigating the impacts of bridge disasters. By leveraging advanced technologies such as IoT, BIM, DT, and AI, this study demonstrates how these phases collectively contribute to the overarching goal of enhancing disaster management efficiency while also identifying areas for future improvement, as shown in Table 5.

Table 5
Key methods/technologies and contributions in the four phases of real-time disaster management.

Phase	Key methods/ technologies	Contribution to Objectives
Pre-Disaster Warning	Sensor networks, IoT, SHM, BIM, ML, AI.	Early risk identification, predictive analytics, and informed decision-making.
Real-Time Disaster Response	UAVs, DT, DISIS, GIS, Sensor Networks.	Rapid damage assessment, emergency decision-making, and efficient resource allocation.
Immediate Post-Disaster Response	TS/dGPS, Computer Vision, ML, AI, BIM, DT.	Comprehensive damage assessment, temporary repair strategies, and secondary disaster prevention.
Post-Disaster Recovery	BIM, DT, Optimised Scheduling & retrofitting.	Long-term rehabilitation, resource management, and transportation network restoration.

5. Real-time bridge disaster management framework (RtBDM)

Current disaster management for bridges requires a framework that integrates real-time monitoring, predictive analytics, and rapid response mechanisms. This section introduces the RtBDM Framework, which is designed to enhance disaster preparedness, response, and recovery through advanced digital technologies. The framework focuses on integrating IoT, digital twins (DT), building information modelling (BIM), artificial intelligence (AI), and machine learning (ML) to create a dynamic and data-driven disaster management approach. This section is structured as follows: Section 5.1 outlines the key requirements for real-time bridge disaster management, emphasising the need for advanced sensing, data processing, and decision-making technologies; Section 5.2 discusses the critical challenges faced in implementing real-time disaster management, including issues related to technology integration, data management, and system coordination; Section 5.3 presents the proposed RtBDM framework, detailing its key components and how it addresses existing gaps in bridge disaster management.

5.1. Requirement analysis

In bridge disaster management, real-time capabilities are critical for saving lives, reducing economic losses, and ensuring the structural integrity of bridges. A real-time disaster management system leverages sensor networks, IoT technology, predictive analytics, and wireless communications to facilitate continuous monitoring and assessment. For instance, accelerometers detect structural vibrations during seismic events, while sonar imaging identifies flood-induced scouring. Thermal cameras monitor fire-related heat distribution, and UAVs assess visible damage during hurricanes. Data collected from these sensors are processed using Finite Element Analysis (FEA) and machine learning models for real-time diagnostics and prognostics. To achieve this, cloud computing platforms integrate multi-source data, and edge computing devices enable rapid localised analysis. Communication protocols such as MQTT and 5G ensure reliable, low-latency data transmission [1]. This integration enables swift responses to potential disasters, helping to prevent or mitigate their effects.

The urgency for real-time disaster management has increased with the global shift toward Industry 5.0, which emphasises the synergy between intelligent technology and human capabilities, aiming for precise monitoring and efficient bridge management before, during, and after disasters [137].

Real-time disaster management represents not only a technological advancement but also an innovation in management concepts. By enabling rapid response and effective intervention at all stages of a disaster, real-time disaster management maximises the protection of human life and property while embodying the trend of modern engineering management toward intelligent and human-centric development. In this background, “real-time” refers to system latency that ensures timely and effective decision-making across different disaster scenarios. For sudden-onset disaster scenarios such as piers or girders deformation detection during earthquakes, latency should ideally be under a few seconds to support immediate safety assessments [138]. For slower-evolving scenarios like flood water level monitoring, delays ranging from a few seconds to a few minutes are generally sufficient for operational decision-making [139]. These benchmarks are proposed based on the practical requirements of disaster management and can be adjusted as per specific case studies or technological advancements.

A critical component of bridge disaster management is the implementation of a comprehensive and accurate early warning system. Traditional systems often rely on limited sensor data and historical disaster models, which limit their ability to integrate and analyse real-time data effectively. For example, while IoT-enabled sensor networks can generate real-time data, challenges arise in correlating this data with historical disaster models to identify potential risks [73]. This limitation hinders the capture and analysis of dynamic changes and potential

threats to bridges. By integrating multi-source data such as bridge physical models, environmental information, real-time data, and historical data, along with advanced algorithms like AI and ML to predict potential disaster patterns, real-time disaster management can issue early warnings at the first indication of potential risks, significantly reducing the likelihood of disasters [82].

Beyond early warning, an efficient emergency response mechanism during disasters is essential. The critical window for rescue and relief is extremely short. Currently, bridge disaster emergency response predominantly relies on manual inspection and static assessment, which lack real-time monitoring and dynamic response capabilities. This results in information asymmetry and irrational resource allocation during post-disaster rescue operations [92]. Therefore, obtaining real-time or near-real-time data to make the optimal decision is critical. The introduction of UAVs, remote sensing, and ground robots for immediate visual assessments and critical data collection, coupled with the dynamic information platform of BIM, can integrate bridge structural data, damage assessment results, and rescue resource information in real time.

Furthermore, scientific and systematic post-disaster recovery planning is essential. Post-disaster recovery should not only focus on repairing the structure itself but also comprehensively consider the overall function and long-term safety of the bridge network. As discussed in Section 4, the primary challenge in this stage is the lack of feedback loops that integrate insights from previous disasters into future planning. For instance, recovery plans often fail to leverage data from the disaster response phase, resulting in missed opportunities for systemic improvements [134]. Effective disaster management requires decision-making across multiple stages (pre-disaster warning, real-time response, immediate post-disaster response, and post-disaster recovery) while connecting condition data and decision-making information acquired at different stages [8]. Constructing a digital management platform that integrates DT models, structural performance indicators, and historical disaster data facilitates the seamless integration of disaster-related data, thereby enhancing comprehensive support for management decisions at every stage. Additionally, the data and lessons learned from current disasters can guide and optimise future disaster response efforts, ultimately forming a closed-loop system of continuous improvement.

However, addressing these needs is not without obstacles. Current technologies often grapple with limitations such as fragmented data management systems, insufficient integration of advanced tools like DT and IoT, and significant resource constraints. These challenges not only hinder the effective implementation of real-time disaster management systems but also underscore the pressing need for further exploration and innovation in the field.

5.2. Critical issues

While significant advancements have been made in the theory and technology of bridge disaster management, numerous challenges persist in practical application, particularly in the areas of technology integration, data management, and system coordination. These challenges highlight the gaps between current technological capabilities and the requirements for effective real-time disaster management. One of the foremost challenges is technology integration. Although technologies such as IoT, DT, BIM, and AI have shown great potential in bridge disaster management and decision-making, they often operate in isolation and lack effective integration and synergy [19,73]. As highlighted in Section 4, while IoT-based sensor networks and DT systems have advanced monitoring and predictive capabilities, their integration with BIM for spatial visualisation or AI for dynamic decision-making remains limited due to a lack of standardisation and interoperability frameworks. For instance, sensor networks provide a vast amount of real-time data, but seamlessly integrating this data with BIM and DT systems for comprehensive analysis and data sharing remains a significant issue [19]. Achieving deep integration of these technologies necessitates advances not only at the technical level but also in standardisation and normalisation, developing a unified integration framework that ensures the flow and application of information across all disaster phases, thereby forming a coherent, information-driven management process.

Another critical challenge is data management. Bridge disaster management must cope with massive, multi-source, and heterogeneous data, which imposes stringent requirements on data collection, transmission, storage, and analysis capabilities. IoT sensors, UAVs, and remote sensing technologies generate diverse data types, including numerical measurements, real-time video feeds, and 3D structural models. However, current systems often fall short in handling real-time and high-frequency data streams effectively [101,102]. Therefore, developing an efficient data management platform that can process, analyse, and store large volumes of data in real time while ensuring data accuracy and completeness remains a key challenge in the field of bridge disaster management. The ability to manage such complex data efficiently is crucial for making informed decisions during a disaster.

Additionally, integration of human-centric and automated systems remains a key obstacle. While Industry 5.0 promotes the adoption of automated decision-making, the reliance on human operators remains crucial in critical disaster scenarios. Balancing these two methods requires addressing the steep learning curve and the significant investment needed for personnel training and system upgrades. Moreover, the fragmented nature of current technologies hinders seamless information flow across disaster phases, exacerbating delays in real-time responses.

Lastly, system coordination is challenging. Bridge disaster management involves multiple departments and institutions, such as traffic management, emergency management, and engineering companies. Achieving efficient coordination among these entities and establishing a unified command and scheduling system is a major obstacle. The current management system is often fragmented, with incomplete information sharing and collaborative working mechanisms, resulting in information silos and resource wastage during disaster response [140]. The construction of a unified command and dispatch platform to facilitate information sharing and collaborative work among various departments is urgently needed to enhance real-time bridge disaster management capability. Dynamic simulation models can further optimise resource allocation and enhance collaboration among stakeholders. This requires not only technological innovation but also institutional and procedural changes to ensure rapid and effective collaboration among all parties during emergencies.

5.3. RtBDM framework

Current bridge management frameworks often cover the entire lifecycle of bridges, particularly those that integrate BIM and DT technologies, covering phases from design and construction to operation and maintenance [141,142]. However, current frameworks lack systematic integration of real-time technologies such as IoT and AI for bridge disaster management. Additionally, they fail to unify BIM and DT into a seamless framework that addresses multi-phase disaster management comprehensively. In response to this gap, Song et al. [12] proposed a DT-enhanced BIM framework specifically for bridge projects, which utilises DT to connect BIM information models from the design phase through construction. This framework supports information management during the operation and maintenance (O&M) phase, facilitating the realisation of a full lifecycle digital bridge project. Despite its comprehensive approach to overall bridge management and optimisation, it does not address the specific needs of bridge disaster response management under disaster conditions, especially when multiple disaster phases are involved.

Building on the BIM&DT framework proposed by Song et al. [12], this paper introduces an enhanced focus on real-time bridge disaster management framework, emphasising rapid response and intelligent decision-making, as illustrated in Fig. 7. Combined with the disaster-defect relationships established in section 3 and utilises the disaster

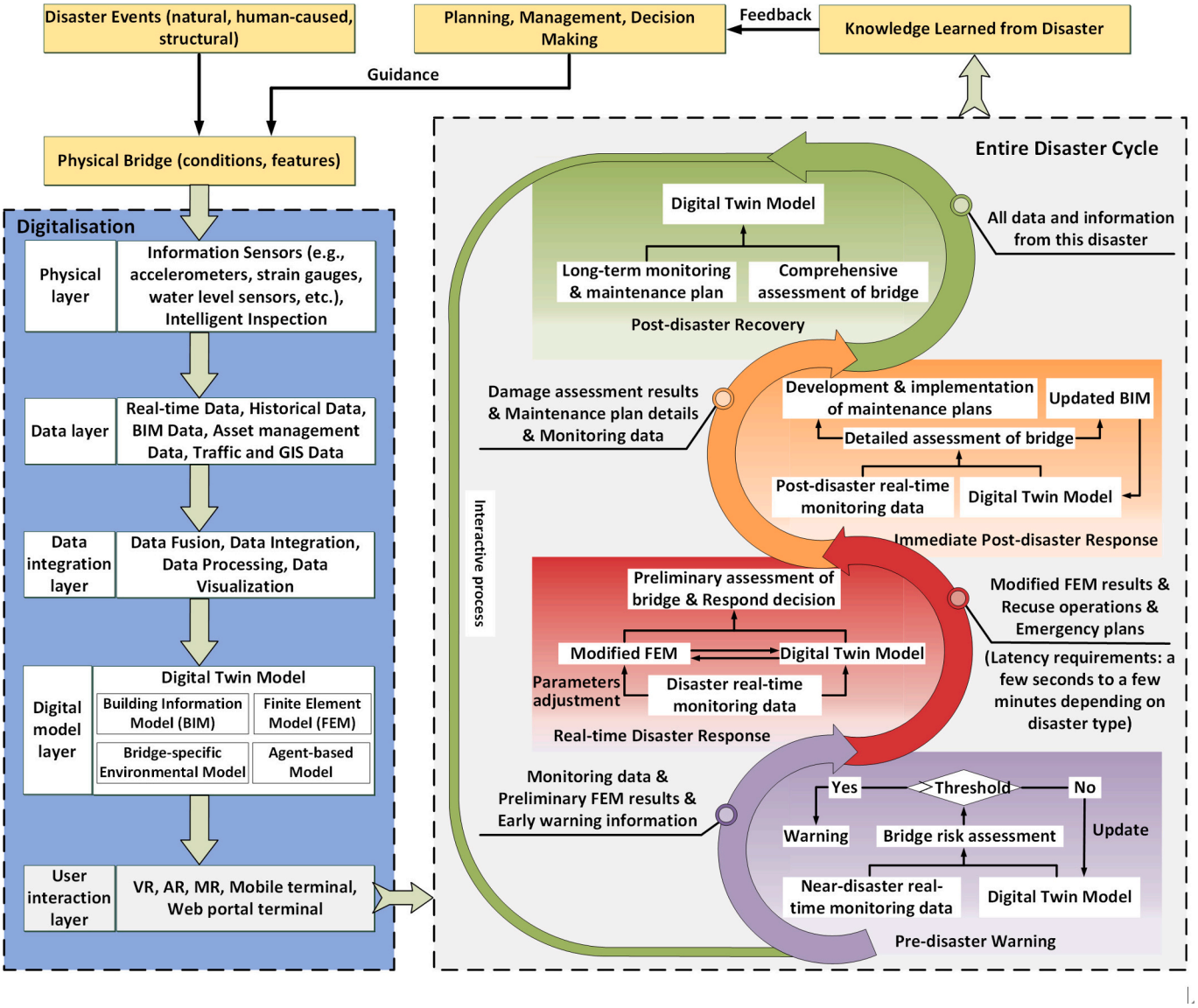


Fig. 7. Real-time bridge disaster management framework.

management phases discussed in section 4, the proposed RtBDM Framework presented a comprehensive system that integrates advanced technologies (such as BIM, DT, IoT, and AI) to provide real-time monitoring, analysis, and decision-making support for bridges before, during, and after disaster events. This enhanced framework addresses critical research gaps by providing dynamic information flow and seamless interaction between integrated systems, enabling proactive disaster response strategies. It proposes an interactive process where feedback from each stage informs the next. Information flows continuously between the digital Twin model, FEM, and other components, ensuring that the system adapts and improves with each disaster event. The proposed framework consists of three key components: (1) Dynamic Adaptation, (2) Digitalisation, and (3) Disaster Cycle. Each of these components plays a crucial role in ensuring the effective management of bridge disasters, from pre-disaster warnings to post-disaster recovery. The components are described as follows:

- **Dynamic Adaptation** (shaded in gold). This component integrates bridge disaster management with disaster events, considering factors such as disaster types, bridge performance status, lessons learned from past disasters, and updated management decisions. Different disasters impact bridge structures in various ways, necessitating adaptive disaster response and management strategies. For instance, in flood scenarios, sensors monitoring water flow, scour, and structural strain are integrated with real-time hydrological models to update risk assessments dynamically. During seismic events, DT models incorporate real-time strain and deformation data to simulate bridge responses, enabling rapid adjustments to mitigation strategies. Bridge performance conditions, including design type, material properties, and current O&M state, influence the performance and response measures of the bridge during a disaster. By analysing and synthesising data and lessons learned from each disaster event, incorporated with real-time monitoring technologies and AI-based predictive analysis, the system dynamically adapts to different disaster scenarios and bridge conditions. This continuous adjustment and upgrading of the disaster management strategies and technical approaches form a closed-loop management system that promotes self-improvement over time.
- **Digitalisation** (shaded in blue). This component is essential for collecting, processing, and integrating data related to the bridge and potential disasters. It consists of five layers: the physical layer, data layer, data integration layer, digital model layer, and user interaction

layer. Those layers work together to create a comprehensive digital representation of the bridge and monitor its status in real-time.

- **Physical layer**. This layer includes sensors and intelligent detection equipment that gather disaster information, such as accelerometers (vibration monitoring), strain gauges (bridge deformation), and water level sensors (flood monitoring). These devices collect real-time data on various parameters such as structural health, environmental conditions, and potential disaster indicators. The data is then transmitted securely using IoT technology. In addition, advanced sensor networks and edge computing are incorporated to enhance the efficiency of real-time data collection and reduce latency.
- **Data layer**. This layer aggregates and stores different types of data. Real-time data captures the current state of the bridge, while historical data provides a reference for past conditions and events. For example, infrared thermal imaging can detect early-stage cracks, while hydrological sensors monitor flood risks in real time. BIM data offers detailed information about the bridge's design and construction, such as structural modelling, while asset management data assists in maintenance planning. Not to mention traffic and GIS data, which provide context on external factors like traffic patterns and geographical conditions.
- **Data integration layer**. In this layer, the collected data undergoes processing. It employs AI, ML, and visualisation tools to filter, clean, integrate, preprocess, and data [143]. This layer also leverages cloud-based platforms for real-time data storage and analysis. The data fusion process combines data from multiple sources to create a full dataset. Following that, data integration enables all data streams to work together seamlessly. This step is then followed by data processing, which involves cleaning and analysing the data, while data visualisation tools present the processed data in an understandable format for decision-makers.
- **Digital model layer.** This layer includes various models such as BIM, finite element simulation model (FEM), environmental model, and agent-based model. Processed data is integrated into the BIM model using BIM software (such as Revit, Rhino, and Bentley) to update the bridge structure information in real-time [11]. Simulation calculations are commonly utilised for structural analysis, damage assessment, safety assessment, and optimisation of bridges under various loads (such as traffic loads, wind loads, and earthquake loads) [27]. The Environmental models simulate various factors in the bridge's environment, including climate conditions, geological conditions, and hydrological characteristics [144]. As a simplified mathematical model, the agent-based model provides rapid prediction and analysis by simulating the input-output relationship of the actual system [145]. The data processed by the data layer updates and optimises the digital model, with big data technology, cloud computing, AI, and ML further supporting data analysis and model prediction. Finally, DT technology is used to integrate these models on a unified platform for real-time monitoring and predictive analysis.
- **User interaction layer.** This layer facilitates interaction between humans, disasters, and digital space through multiple terminals, such as VR, AR, MR, mobile terminals, and web portal terminals [146]. This allows stakeholders to visualise and interact with the bridge's digital twin, making it easier to monitor conditions and make informed decisions. The bridge disaster management stage focused on in this article relies on the services provided by this layer. The digital component offers a real-time dynamic update platform for bridge disaster management, supporting pre-disaster warning, disaster response, disaster recovery, and long-term maintenance.

- **Disaster cycle** (shaded in grey). This component encompasses four key subcomponents: pre-disaster warning, real-time disaster response, immediate post-disaster response, and post-disaster recovery. The workflow of each component is as follows:
  - **Pre-Disaster Warning (Purple).** Utilising the digital twin model and real-time monitoring data, the system continuously monitors structural anomalies and conducts bridge disaster risk assessments. Key performance metrics, such as deflection, crack extension, spalling area, rebar exposure, and pier deformation, are evaluated and an early warning is issued if the assessment result exceeds the index threshold. The structural analysis results and early warning information are then passed to the next stage.
  - **Real-Time Disaster Response (Red)**. During a disaster, real-time data is collected to assess the bridge's condition using the bridge structure's basic information provided by BIM and the real-time data collected at the disaster site. This data is fed into the FEM, reflecting on the current stress distribution and deformation of the bridge. The information is fed back to the DT model to quickly assess the bridge's status and on-site damage, supporting immediate emergency decision-making and response measures. The updated FEM results and emergency measures obtained during the real-time disaster response stage provide reference data for subsequent responses.
  - **Immediate Post-Disaster Response (Orange)**. After the disaster has subsided, collected data, image information, and model analysis results are used to conduct a detailed assessment of bridge damage. This assessment helps determine structural repair priorities and plans, guiding emergency repair and reinforcement work. The digital model is then updated to reflect the latest bridge and disaster management status to refine long-term recovery plans.
  - **Post-Disaster Recovery (Green)**. Based on the detailed damage assessment and repair results, the framework further refines the bridge restoration process, formulates a long-term monitoring and maintenance plan, and feeds back into the DT model. All data and information generated by the disaster are looped back to the pre-disaster warning stage, forming a continuous iteration of information. The experience and knowledge gained from managing the disaster are used to update further management decisions and provide guidance for handling similar scenarios on other bridges.

  Unlike generic disaster management systems, the RTBDM framework is tailored specifically to address challenges unique to bridges:

1) Employs accelerometers, strain sensors, and displacement transducers to assess dynamic responses and cumulative fatigue effects caused by earthquakes, traffic loads, or extreme weather conditions in critical bridge components, such as girders, bearings, and piers.
2) Integrates data form hydrological sensors, geotechnical monitoring systems, and multi-source imaging techniques to provide comprehensive insights during flood or landslide events, enhancing predictive capabilities for cascading disasters.
3) Incorporates bridge information system, SHM system, or other inspection applications to detect and quantify concrete spalling, cracks, and deformation caused by environmental exposure, heavy loads, or sudden impacts.

6. Conclusion

This paper provided a systematic overview of the current research landscape in bridge disaster management, addressing the critical phases of pre-disaster warning, real-time disaster response, immediate post-disaster response, and post-disaster recovery. By identifying key challenges, such as the integration of heterogeneous data sources and the lack of robust decision-making frameworks, the paper emphasises the urgent need for real-time and intelligent disaster management solutions.

In response, the proposed RtBDM framework offers a cohesive and dynamic approach to addressing these challenges. The framework is built on three main components: dynamic adaptation, digitalisation, and disaster cycle. The dynamic adaptation component enables real-time adjustments to disaster strategies by integrating disaster types, bridge

information, and management decisions. The digitalisation component integrates BIM and DT technologies to enable full lifecycle management of bridge data and provide intelligent disaster response solutions through automated decision-making systems. The disaster cycle covers the entire management process, from pre-disaster warning and real-time disaster response to post-disaster recovery, ensuring seamless connectivity and information interoperability at different stages. At its core, the RtBDM framework integrates advanced sensor technologies, big data analytics, IoT, and artificial intelligence to facilitate real-time monitoring, intelligent early warnings, rapid response capabilities, and efficient recovery processes for bridge disasters.

Despite its contributions, the study acknowledges certain limitations. The analysis primarily depended on existing research and the available technological tools, potentially limiting the exploration of emerging innovations. Furthermore, the proposed framework may require ongoing updates to remain aligned with technological advancements. Additionally, the framework's effectiveness in various real-world scenarios has yet to be fully validated. Addressing these gaps will require extensive pilot implementations and case studies across different regions, disaster types, and bridge systems to refine and optimise the framework further. Future research should focus on further enhancing the application of the RtBDM framework, particularly in improving technology integration and refining data management, to promote the dynamic and intelligent development of bridge disaster management. The RtBDM framework not only advances the state of bridge disaster management but also provides a comprehensive methodology that can be applied to other critical infrastructure systems.

CRediT authorship contribution statement

Hairong Deng: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Haijiang Li:** Writing – review & editing, Supervision, Conceptualization. **Lueqin Xu:** Writing – review & editing, Supervision, Funding acquisition. **Ali Khudhair:** Writing – review & editing, Visualization, Methodology. **Honghong Song:** Visualization. **Yu Gao:** Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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